

- 7 (a) nuclei/atoms with same proton number/atomic number B1
 nuclei/atoms contain different numbers of neutrons/different atomic mass B1 [2]
- (b) (i) 92 A1 [1]
 (ii) 146 A1 [1]
- (c) (i) mass = $238 \times 1.66 \times 10^{-27}$ C1
 = 3.95×10^{-25} kg A1 [2]
- (ii) volume = $\frac{4}{3} \pi \times (8.9 \times 10^{-15})^3$ (= 2.95×10^{-42}) C1
 density = $(3.95 \times 10^{-25}) / (2.95 \times 10^{-42})$
 = 1.3×10^{17} kg m⁻³ A1 [2]
- (d) nucleus contains most of mass of atom B1
 either nuclear diameter/volume very much less than that of atom
 or atom is mostly (empty) space B1 [2]